

Day 19: Amazon S3 Integration with EC2 Using IAM Roles, AWS CLI & S3 Mounting

Learning Objectives

Understand Amazon S3 concepts

- Create and configure S3 buckets
 - Upload and manage objects in S3
 - Attach IAM Roles to EC2 instances
 - Access S3 from EC2 using AWS CLI
 - Transfer files between EC2 and S3
 - Mount an S3 bucket on an EC2 instance
 - Understand differences between EBS, EFS, and S3
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1. What is Amazon S3?

Amazon S3 (Simple Storage Service) is an object storage service used to store and retrieve data from anywhere on the internet.

Features

- Highly Scalable
- Secure
- Durable (99.999999999%)
- Cost Effective
- Unlimited Storage Capacity

Common Use Cases

- Backup and Recovery
- Static Website Hosting
- Log Storage
- Media Storage
- Application Data Storage

2. S3 Terminology

Bucket

A container used to store files.

Example:

my-devops-bucket

Object

A file stored inside a bucket.

Examples:

test.txt

image.png

backup.zip

Key

Unique path of an object.

Example:

documents/test.txt

3. Create an S3 Bucket

Step 1

Login to AWS Console

Step 2

Navigate to:

S3 → Create Bucket

Step 3

Provide:

Bucket Name: cloudservicessai

Region: us-east-1

Keep all other settings as default.

Step 4

Click:

Create Bucket

4. Upload Files to S3

Open the bucket.

Click:

Upload

Select:

test.txt

Click:

Upload

5. Launch an EC2 Instance

Launch an Ubuntu or Amazon Linux instance.

Connect to EC2:

Ubuntu:

```
ssh -i key.pem ubuntu@public-ip
```

Amazon Linux:

```
ssh -i key.pem ec2-user@public-ip
```

6. Install AWS CLI

Check AWS CLI:

```
aws --version
```

Ubuntu:

```
sudo apt update
sudo apt install awscli -y
```

Amazon Linux:

```
sudo yum install awscli -y
```

Verify:

```
aws --version
```

7. Create IAM Role for EC2

Why IAM Role?

IAM Roles allow EC2 instances to access AWS services securely without storing access keys.

Steps

Navigate to:

IAM → Roles → Create Role

Choose:

AWS Service

Select:

EC2

Attach Policy:

AmazonS3FullAccess

Role Name:

EC2-S3-Role

Click:

Create Role

8. Attach IAM Role to EC2

Navigate:

EC2 → Instances

Select your instance.

Choose:

Actions → Security → Modify IAM Role

Select:

EC2-S3-Role

Save.

9. Verify S3 Access from EC2

Check identity:

```
aws sts get-caller-identity
```

List buckets:

```
aws s3 ls
```

Expected Output:

```
cloudservicessai
```

10. Upload File from EC2 to S3

Create a file:

```
echo "Hello AWS S3" > test.txt
```

Upload:

```
aws s3 cp test.txt s3://cloudservicessai/
```

Verify:

```
aws s3 ls s3://cloudservicessai/
```

11. Download File from S3 to EC2

```
aws s3 cp s3://cloudservicessai/test.txt .
```

Verify:

```
cat test.txt
```

12. Sync Directory to S3

Create directory:

```
mkdir backup
```

```
touch backup/file1
```

```
touch backup/file2
```

Sync:

```
aws s3 sync backup/ s3://cloudservicessai/backup/
```

13. Sync S3 to EC2

```
aws s3 sync s3://cloudservicessai/backup/ backup/
```

14. Common S3 Commands

List Buckets:

```
aws s3 ls
```

List Objects:

```
aws s3 ls s3://cloudservicessai
```

Upload File:

```
aws s3 cp file.txt s3://cloudservicessai/
```

Delete File:

```
aws s3 rm s3://cloudservicessai/file.txt
```

Delete Bucket:

```
aws s3 rb s3://cloudservicessai --force
```

15. Mount S3 Bucket to EC2

Important

S3 is an Object Storage service.

Unlike EBS and EFS, S3 cannot be mounted directly.

To mount S3, use:

- Mountpoint for Amazon S3 (Recommended)
 - s3fs-fuse
-

Method 1: Mountpoint for Amazon S3

Verify IAM Role

```
aws sts get-caller-identity
```

```
aws s3 ls
```

Install Mountpoint

Ubuntu:

```
wget https://s3.amazonaws.com/mountpoint-s3-release/latest/x86_64/mount-s3.deb
```

```
sudo apt install ./mount-s3.deb -y
```

Verify:

```
mount-s3 --version
```

Create Mount Directory

```
sudo mkdir /s3data
```

Mount Bucket

```
sudo mount-s3 cloudserviceai /s3data
```

Verify

```
df -h
```

```
ls -l /s3data
```

Create a test file:

```
echo "Hello S3" > /s3data/test.txt
```

Verify in S3:

```
aws s3 ls s3://cloudservicessai/
```

Method 2: Using s3fs-fuse

Install:

```
sudo apt update  
sudo apt install s3fs -y
```

Create Mount Point:

```
sudo mkdir /s3bucket
```

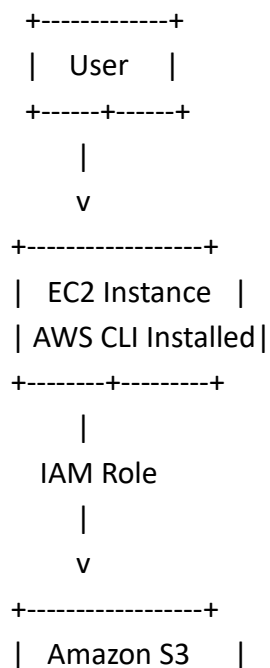
Mount:

```
sudo s3fs cloudservicessai /s3bucket \  
-o iam_role=auto \  
-o allow_other
```

Verify:

```
df -h  
ls /s3bucket
```

Architecture Diagram



| cloudfactory |
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(Optional Mount using Mountpoint/S3FS)

Difference Between EBS, EFS and S3

Feature	EBS	EFS	S3
Storage Type	Block	File	Object
Mount Directly	Yes	Yes	No
Multi-Instance Access	No	Yes	Via Tools
Use Case	OS, Database Shared Storage Backup, Logs, Static Content		

Interview Questions

Q1. What is Amazon S3?

Amazon S3 is an object storage service used to store and retrieve files.

Q2. What is a Bucket?

A bucket is a container used to store objects in S3.

Q3. What is an Object?

An object is a file stored inside an S3 bucket.

Q4. Why use IAM Roles instead of Access Keys?

IAM Roles provide secure access without hardcoding credentials.

Q5. Which command uploads files to S3?

```
aws s3 cp file.txt s3://bucket-name/
```

Q6. Which command synchronizes data?

```
aws s3 sync source destination
```

Q7. Can S3 be mounted directly like EBS or EFS?

No. S3 is object storage and requires Mountpoint for Amazon S3 or s3fs-fuse.

Q8. Difference between EBS and S3?

- EBS is Block Storage.
- S3 is Object Storage.

Real-Time Lab Demo

1. Create an S3 Bucket.
2. Launch an EC2 Instance.
3. Create an IAM Role.
4. Attach IAM Role to EC2.
5. Verify S3 access using AWS CLI.
6. Upload a file from EC2 to S3.
7. Download a file from S3 to EC2.
8. Perform S3 Sync operations.
9. Mount S3 bucket using Mountpoint for Amazon S3.
10. Explain differences between EBS, EFS, and S3.